

GlucoGuide: A Retrospective Counterfactual Decision Support to Address Stakeholder Needs

Asiful Arefeen, Shovito Barua Soumma, Md. Nazmul Islam, and Hassan Ghasemzadeh

Abstract—Counterfactual explanations (CFs) refer to minimal alterations to user attributes fed into a model that result in a better predicted outcome. Although research on CFs has existed for years, little effort has been made to apply them to practical problems faced by stakeholders in the health sector. We present *GlucoGuide*, a software tool designed to assist endocrinologists in making retrospective clinical decisions when consulting with patients with Type 1 Diabetes (T1D) who use Automated Insulin Delivery (AID) and Continuous Glucose Monitoring (CGM) technology. GlucoGuide can help patients with T1D understand how they could have improved insulin dosing adjustments to achieve better glycemic control in situations where they experienced hyperglycemia. GlucoGuide consists of a classifier, a CF generator as well as a forecaster, which can be used to verify if the interventions can keep the resulting glycemic response within good range and prevent hyperglycemia. With all these components, GlucoGuide aims to provide an end-to-end decision support for retrospective analysis and clinical guidance through insulin adjustment.

Index Terms—Counterfactual explanations, Decision Support Systems, Diabetes, Digital health, Explainable AI, Metabolic health, Wearable sensors

I. GLUCOGUIDE

GlucoGuide is meant to be used in a clinical setting. Patients with T1D will visit their care providers, who will load the data from patient’s AID device, visualize the data and make retrospective recommendations on cases where patients could have done better. To develop GlucoGuide, we have collected data from 38 patients monitored for 26 consecutive days each while using Tandem T:SLIM X2 Insulin Pump and Dexcom G6 Pro CGM sensor in free-living condition. The data contains continuous blood glucose levels (BGL) sampled at 5 minutes, insulin logs and their corresponding carb sizes.

The first step is to **process** the data. Since the best way to communicate interventions with end-users is using tabular data, the continuous data from the insulin pump is transformed into tabular format. The context is a 3-hour long window, while each model in GlucoGuide, whether it predicts hyperglycemia or forecasts BGL, looks 90 minutes into the future. The 3-hour long context is segmented into six isochronal bins, first

five of which are identical, containing total carb consumed and total bolus taken within the 30 minutes. The last bin (bin # 6) contains granular information regarding the carb amount, bolus amounts of different types and their timings relative to the end of the window. Patient demographics, slope of BGL (last 30 minutes) and last BGL reading of the window is also added to the context. Two consecutive windows are 15 minutes apart (i.e., approx 92% overlap in context). The label is set as hyperglycemic (≥ 180 mg/dL) or normoglycemic based on the BGL readings in 90 minute window.

Using the processed data, a Dual Attention Network (DANet) **classifier** is trained that can detect hyperglycemia beforehand. Additionally, a conditional Variational Autoencoder (cVAE) **generator** is trained to generate CF recommendations. When fed with a context that leads to hyperglycemia, the role of the generator is to generate a CF recommendation such that the classifier predicts it as a normoglycemic sample. The generator only alters some of the contents of bin # 6 (insulin amounts and timings) and last BGL reading of the window to flip the class, other attributes remain unchanged.

The last component is a **forecaster**, built with a sequence-to-sequence transformer. The forecaster helps providers verify if the recommendations lead to positive outcomes within the desired glucose range (Figure 2). Furthermore, providers can change context attributes and design interventions themselves and visualize outcomes using the forecaster. The performance of GlucoGuide is summarized in Table I.

TABLE I: Evaluating individual components of GlucoGuide.

Classifier Performance		
Accuracy (%)	F1-score	AUC
84.12%	83.91%	0.9063
CF Generator Performance		
Validity (%)	L_2 Distance	Plausibility (%)
94.29%	63.86	98.22%
Forecaster Performance		
RMSE (mg/dL)	MAE (mg/dL)	R^2
7.24	5.02	0.976

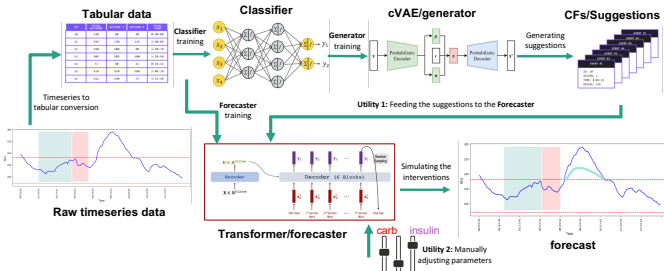


Fig. 1: Working pipeline of GlucoGuide including all components (data processing, classifier, CF generator, and forecaster).

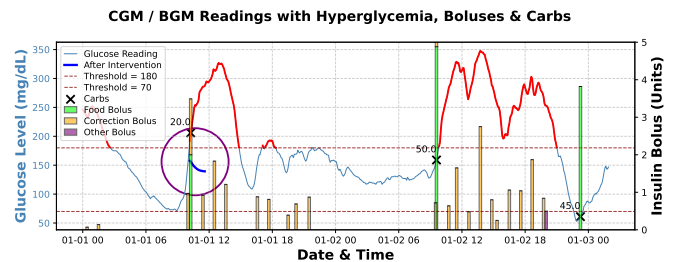


Fig. 2: Visualizing intervention outcomes using GlucoGuide.